IGCSE 0607 Practice Questions — E4.6 — Combined Questions & Markscheme

Perpendicular lines

Total Questions: 6 — **Total Marks:** 36 Marks

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  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(2 Qs)*

1 Line L_1 has the equation $y = 5 - 2x$.
Line L_2 is perpendicular to L_1 and passes through the point $(4, 1)$.

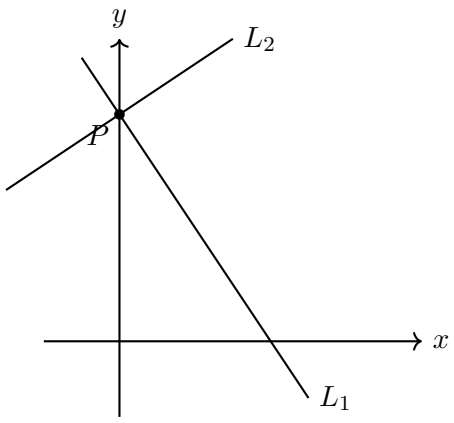
(a) Find the gradient of line L_1.

..... **[1]**

(b) Find the equation of line L_2.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

2 Line L_1 has the equation $3x + 2y = 12$.
Line L_2 is perpendicular to L_1 and intersects L_1 at a point P on the y-axis.



Not to scale

(a) Find the coordinates of point P.

(..... , ...) **[1]**

(b) Find the equation of line L_2.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

(c) Point Q lies on L_2 and has an x-coordinate of 9.
Point R lies on L_1 and has an x-coordinate of 4.
Work out the area of triangle PQR.

..... **[5]**

3 Find the equation of the line that is perpendicular to $y = 3x - 5$ and passes through the point $(6, 2)$. Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

4 A is the point $(-4, 7)$ and B is the point $(2, -1)$.

Find the equation of the perpendicular bisector of the line segment AB.
Give your answer in the form $ax + by = c$, where a, b and c are integers.

..... **[5]**

5 The points A and B have coordinates $(-2, 4)$ and $(6, 0)$ respectively.

(a) Find the coordinates of the midpoint of AB.

(..... , ...) **[2]**

(b) Find the equation of the perpendicular bisector of the line segment AB.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

6 The line L_1 has equation $2y + 3x = 12$.
The line L_2 is perpendicular to L_1 and intersects L_1 at the point P on the y-axis.

(a) Find the coordinates of P.

(..... , ...) **[2]**

(b) Find the equation of line L_2.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

(c) Line L_2 crosses the x-axis at the point Q.
Calculate the exact length of PQ.

..... **[4]**

1 **(a)** -2 **[B1]**

(b) Gradient of $L_2 = \frac{1}{2}$ **[M1]**

$1 = \frac{1}{2}(4) + c$ or $y - 1 = \frac{1}{2}(x - 4)$ **[M1]**

$y = \frac{1}{2}x - 1$ **[A1]**

[Back to Question] *[Total Marks: 4]*

2 **(a)** $(0, 6)$ **[B1]**

(b) Gradient of $L_1 = -\frac{3}{2}$ **[M1]**

Gradient of $L_2 = \frac{2}{3}$ **[M1]**

$y = \frac{2}{3}x + 6$ **[A1]**

(c) Coordinates of Q: $y = \frac{2}{3}(9) + 6 = 12 \implies Q(9, 12)$ **[B1]**

Coordinates of R: $3(4) + 2y = 12 \implies y = 0 \implies R(4, 0)$ **[B1]**

Identify triangle PQR is right-angled at P OR setup valid box/shoelace method **[M1]**

Length $PQ = \sqrt{9^2 + 6^2} = \sqrt{117}$ and Length $PR = \sqrt{4^2 + (-6)^2} = \sqrt{52}$ **[M1]**

Area $= \frac{1}{2} \times \sqrt{117} \times \sqrt{52} = 39$ **[A1]**

[Back to Question] *[Total Marks: 9]*

3 Identifies the gradient of the given line as 3 **[B1]**

Uses the perpendicular gradient $-\frac{1}{3}$ and substitutes $(6, 2)$ into $y = mx + c$ or $y - y_1 = m(x - x_1)$ **[M1]**

$-\frac{1}{3}x + 4$ **[A1]**

[Back to Question] *[Total Marks: 3]*

4 Calculates the midpoint of $AB = \left(\frac{-4+2}{2}, \frac{7+(-1)}{2}\right) = (-1, 3)$ **[B1]**

Calculates the gradient of $AB = \frac{-1-7}{2-(-4)} = -\frac{8}{6} = -\frac{4}{3}$ **[B1]**

Finds the perpendicular gradient $= \frac{3}{4}$ **[M1]**

Substitutes the midpoint $(-1, 3)$ and gradient $\frac{3}{4}$ to find the equation, e.g., $y - 3 = \frac{3}{4}(x - (-1))$ **[M1]**

$3x - 4y = -15$ or $4y - 3x = 15$ or equivalent integer form **[A1]**

[Back to Question] *[Total Marks: 5]*

5 **(a)** $\left(\frac{-2+6}{2}, \frac{4+0}{2}\right)$ **[M1]**

$(2, 2)$ **[A1]**

(b) Gradient of $AB = \frac{0-4}{6-(-2)} = -\frac{1}{2}$ **[M1]**

Gradient of perpendicular bisector $= 2$ **[M1]**

Substitute midpoint $(2, 2)$ into $y - y_1 = m(x - x_1)$: $y - 2 = 2(x - 2)$ **[M1]**

$y = 2x - 2$ **[A1]**

[Back to Question] *[Total Marks: 6]*

6 **(a)** Substitutes $x = 0$ into $2y + 3x = 12$ because P is on the y-axis **[M1]**

$(0, 6)$ **[A1]**

(b) Rearranges L_1 equation to find gradient: $2y = -3x + 12 \implies y = -\frac{3}{2}x + 6$, so $m = -\frac{3}{2}$ **[B1]**

States perpendicular gradient for L_2 is $\frac{2}{3}$ **[M1]**

$\frac{2}{3}x + 6$ **[A1]**

(c) Substitutes $y = 0$ into the equation of L_2 to find the coordinates of Q **[M1]**

$0 = \frac{2}{3}x + 6 \implies x = -9$ **[A1]**

Uses the distance formula for $P(0, 6)$ and $Q(-9, 0)$: $\sqrt{(-9-0)^2 + (0-6)^2}$ **[M1]**

$\sqrt{117}$ or $3\sqrt{13}$ **[A1]**

[Back to Question] *[Total Marks: 9]*